

Summary Data on Acid Cropland and Lime Requirements in Rwanda

GAIA

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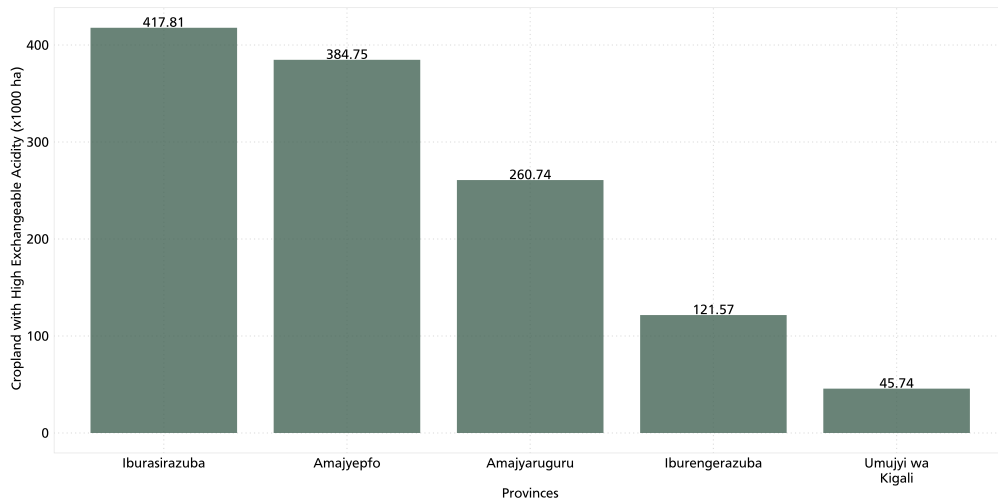
Outlook

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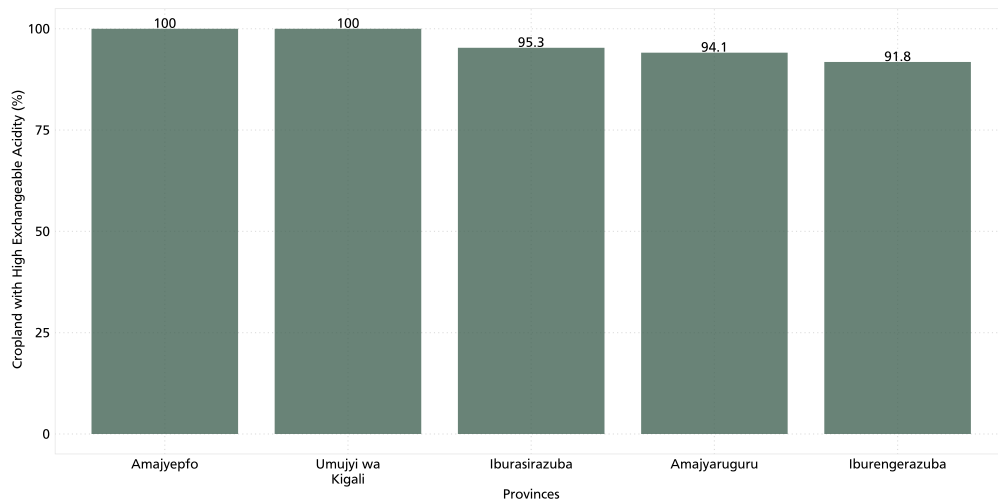
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Rwanda**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production and production value from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

Top 10 Provinces in Rwanda with the highest acidic cropland



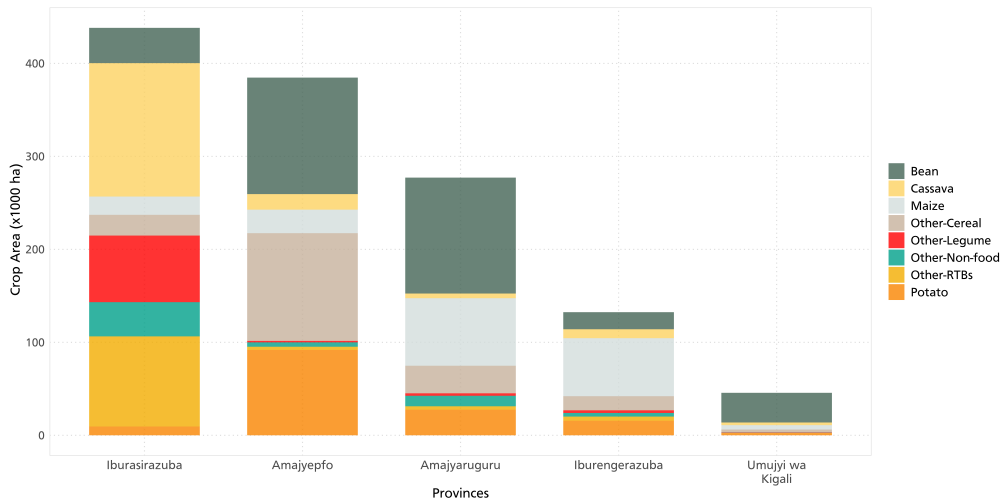
Top 10 Provinces in Rwanda with the highest acidic cropland (Percentage of total cropland)



└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland



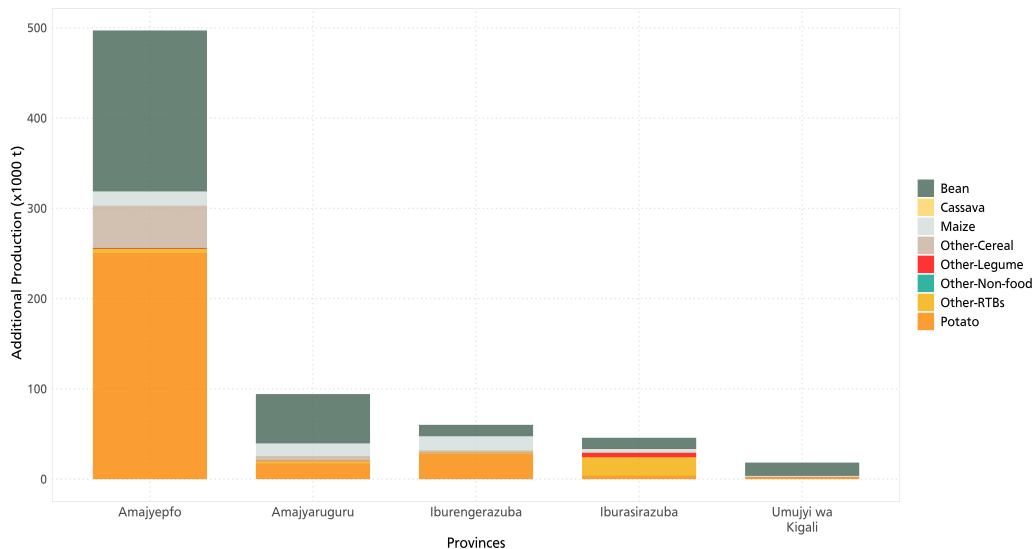
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland (table)

Provinces	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	124.66	5.03	72.61	29.67	2.67	11.32	3.68	27.53
Amajyepfo	125.33	16.69	25.34	115.82	1.59	4.70	3.50	91.79
Iburasirazuba	37.96	143.45	19.67	22.34	71.59	36.79	97.00	9.47
Iburengerazuba	18.36	9.56	62.46	15.18	2.88	3.87	4.47	15.66
Umujyi wa Kigali	31.90	2.93	4.72	3.03	0.36	0.26	0.16	2.39

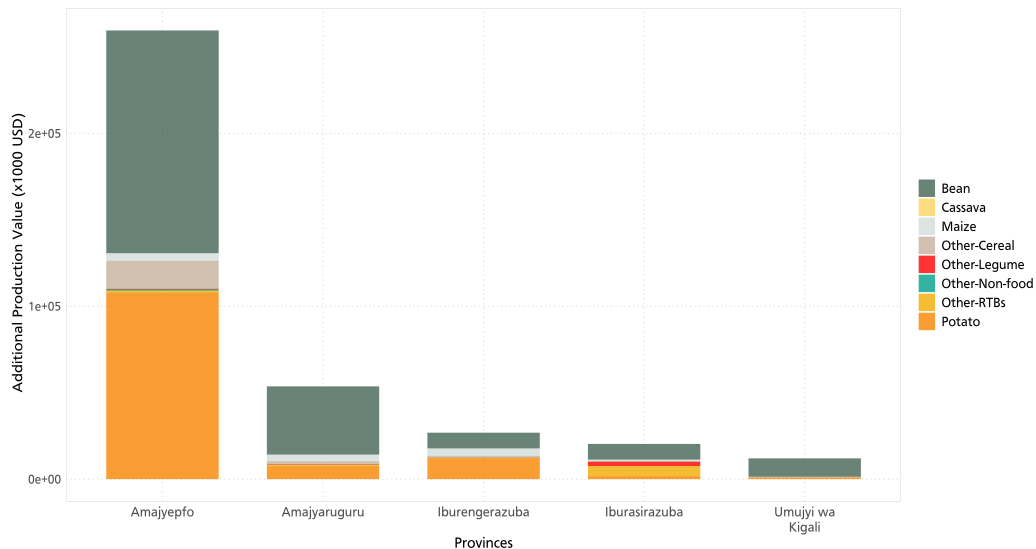
Potential for additional production (tone) in top 10 acidic Provinces in Rwanda



Potential for additional production (1000 tone) in top 10 acidic Provinces in Rwanda(Table)

Provinces	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	54.63	-	13.90	5.27	0.43	0.02	2.62	17.38
Amajyepfo	178.29	0.07	15.77	46.76	0.70	0.29	4.54	250.67
Iburasirazuba	12.49	0.05	3.11	1.16	4.69	0.04	20.42	3.91
Iburengerazuba	12.60	-	15.63	2.76	0.19	0.05	0.97	28.00
Umujyi wa Kigali	14.74	0.08	0.83	0.44	0.11	-	0.14	2.03

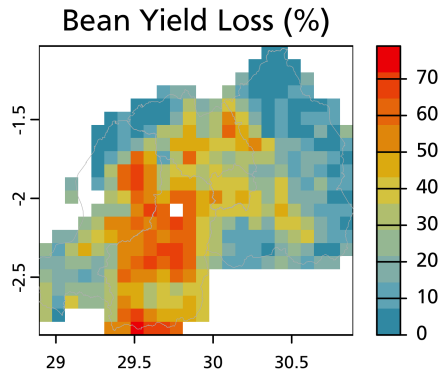
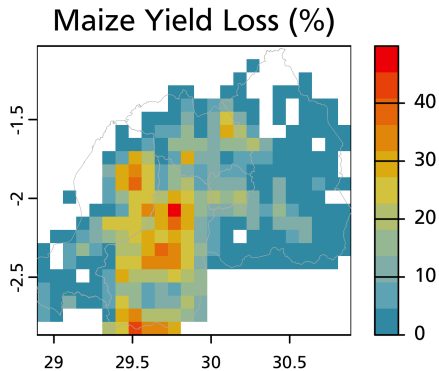
Potential for additional production value (US\$) in top 10 acidic Provinces in Rwanda



Potential for additional production value (1000 US\$) in top 10 acidic Provinces in Rwanda(Table)

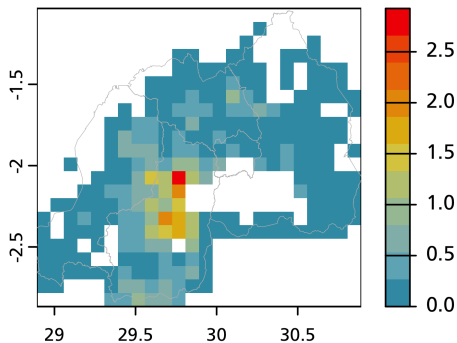
Provinces	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	39463.0	0.3	3861.2	1875.2	233.8	26.5	761.6	7468.8
Amajyepfo	128799.0	18.8	4380.2	16231.0	347.8	684.6	1318.9	107748.6
Iburasirazuba	9019.2	13.7	863.8	403.5	2419.7	31.6	5936.1	1679.9
Iburengerazuba	9103.1	-	4342.4	991.2	101.6	47.6	280.7	12035.4
Umujiyi wa Kigali	10651.3	20.2	230.6	147.8	65.1	2.9	40.6	872.3

Yield loss due to soil acidity for Maize and Bean in Rwanda

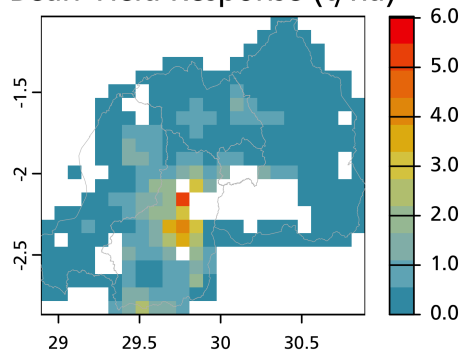


Yield response to lime application in Rwanda

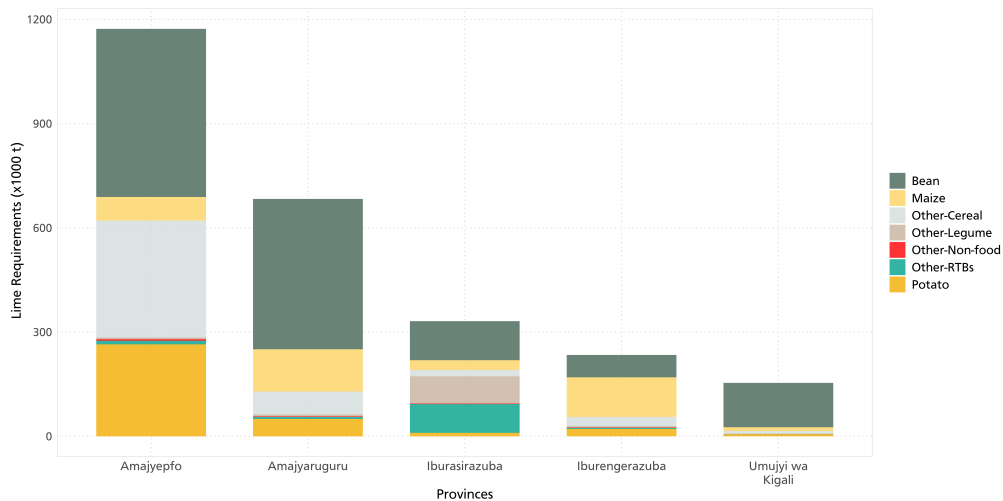
Maize Yield Response (t/ha)



Bean Yield Response (t/ha)



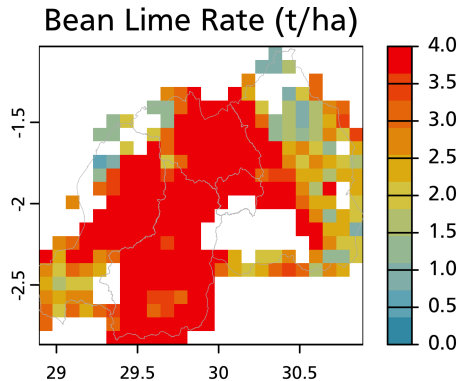
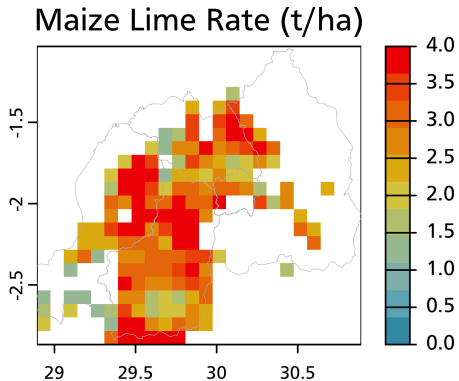
Lime requirements (t) for acid soil remediation in Rwanda



Lime requirements (1000 t) for acid soil remediation in Rwanda (table)

Provinces	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	432.84	121.89	65.42	5.77	1.92	5.64	49.94
Amajyepfo	483.80	67.90	336.53	5.00	4.56	10.30	264.71
Iburasirazuba	112.37	27.81	18.61	78.06	1.56	82.91	10.07
Iburengerazuba	64.34	114.04	26.35	3.81	1.22	3.23	21.18
Umujyi wa Kigali	127.61	10.90	7.65	1.31	0.03	0.42	5.97

Lime rate for Maize and Bean in Rwanda



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.